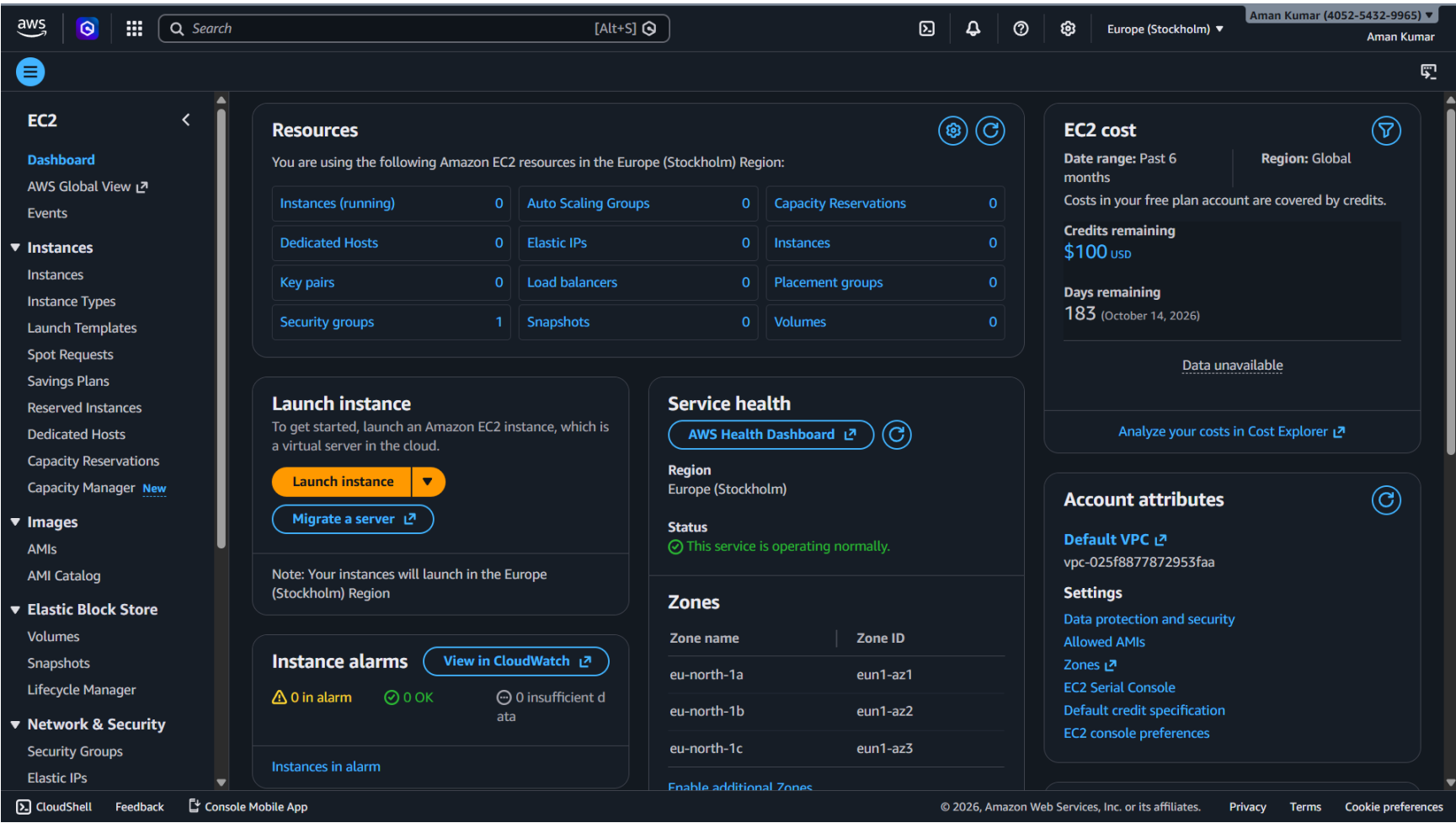
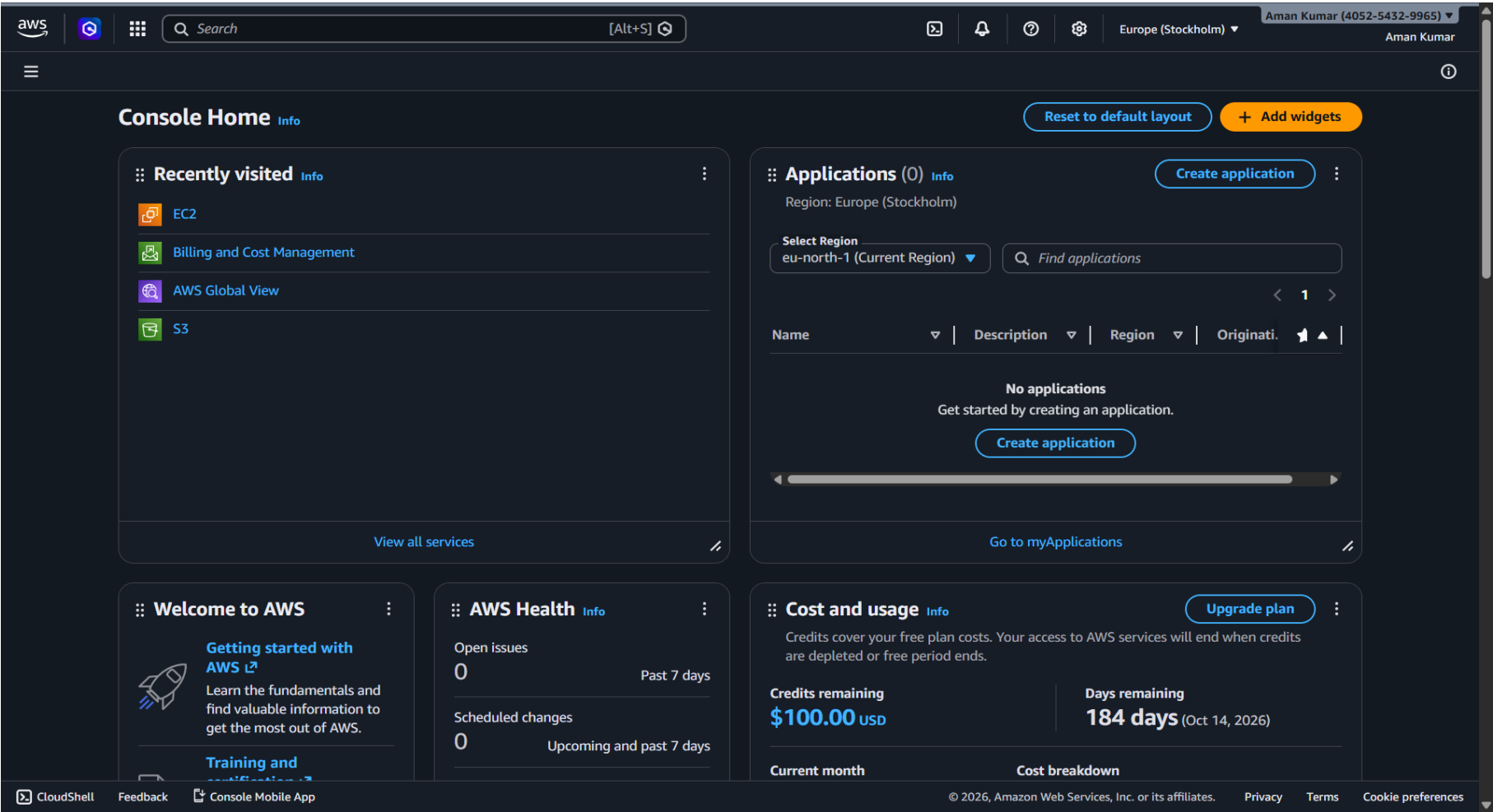


PRACTICAL 1:

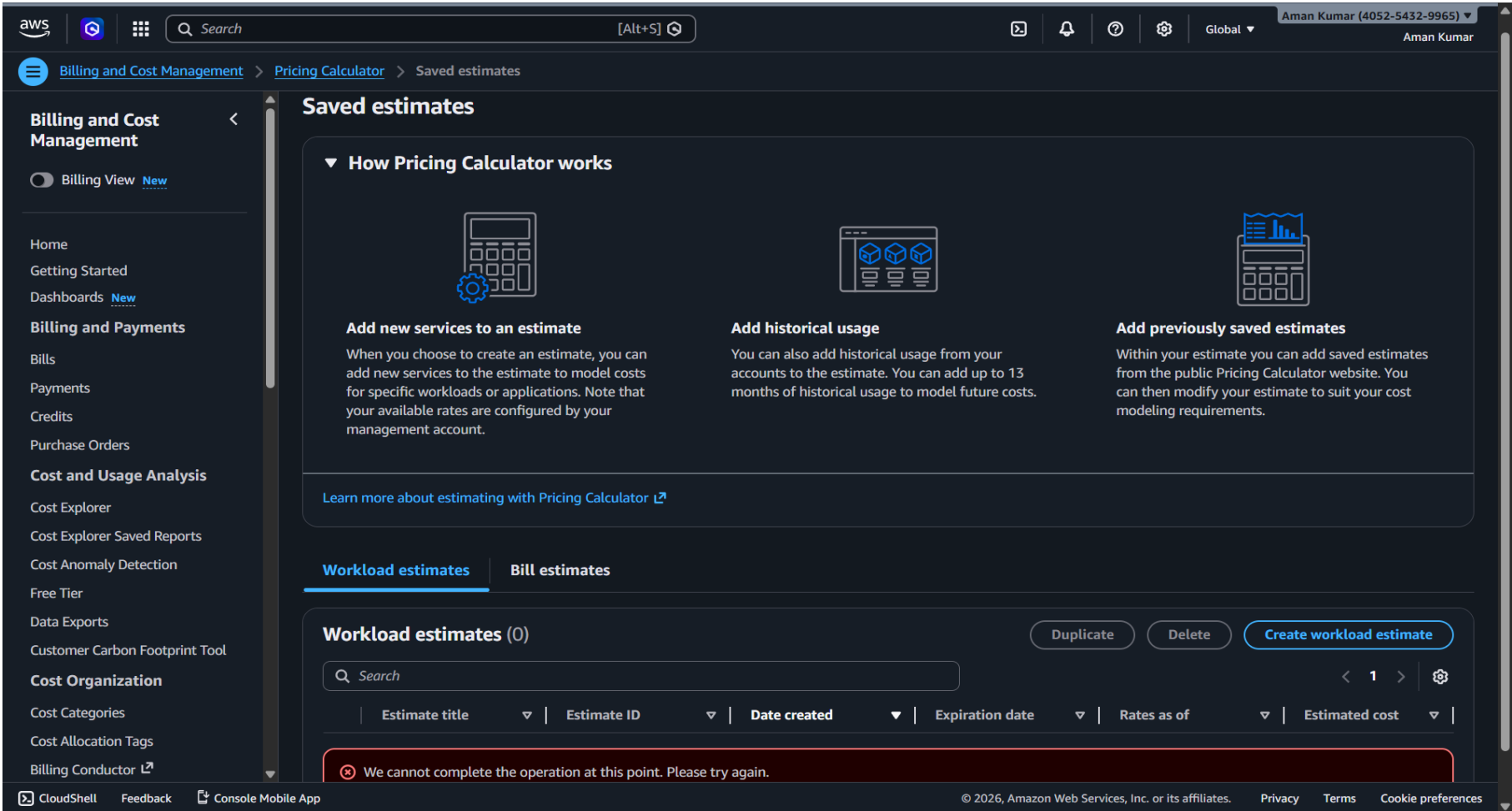
Aim: To familiarize with cloud platforms and instances.

Procedure:

1. Create an account on AWS.
2. Explore dashboard and identify key services.



3. Pricing Calculator.

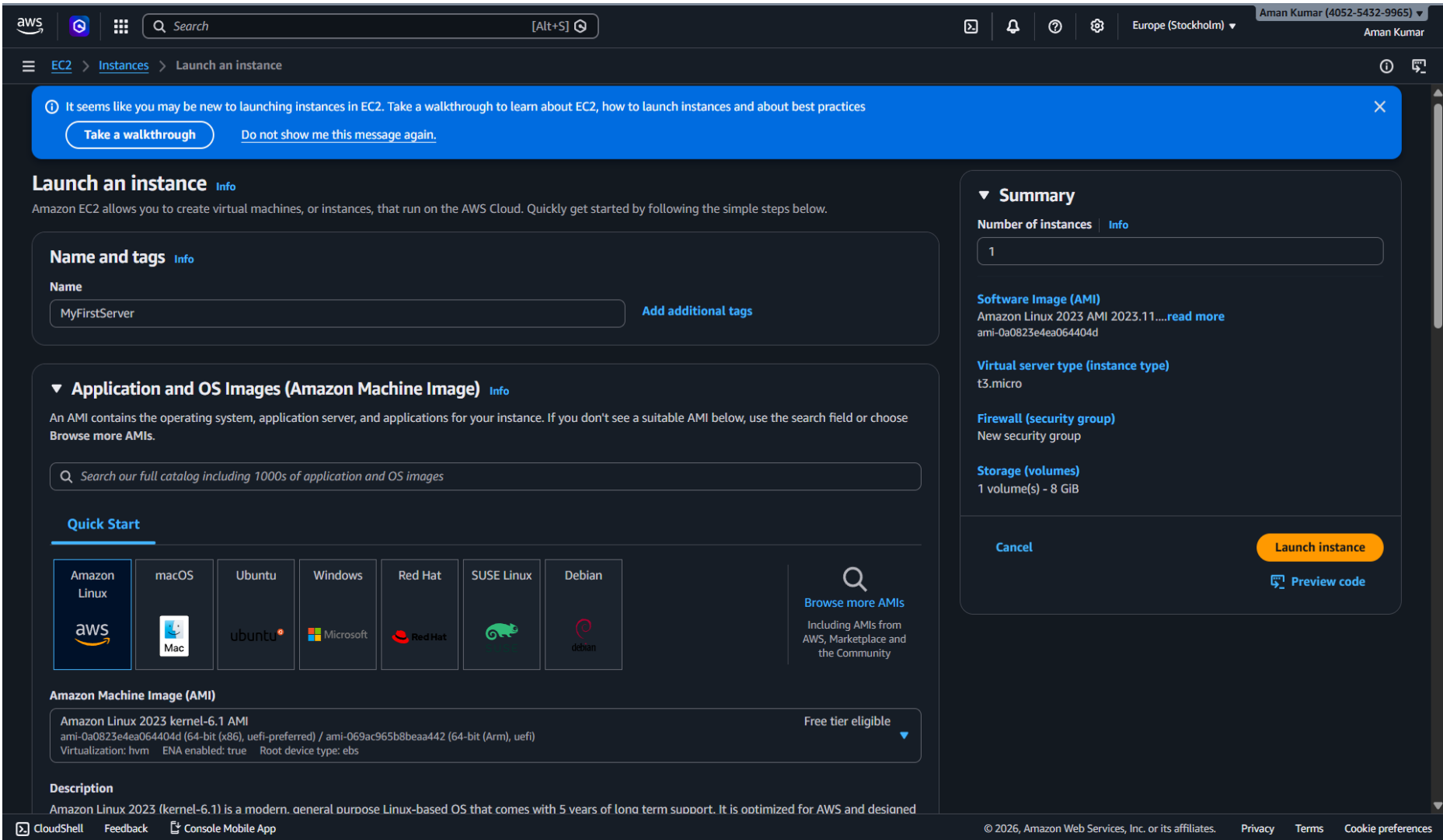


PRACTICAL 2:

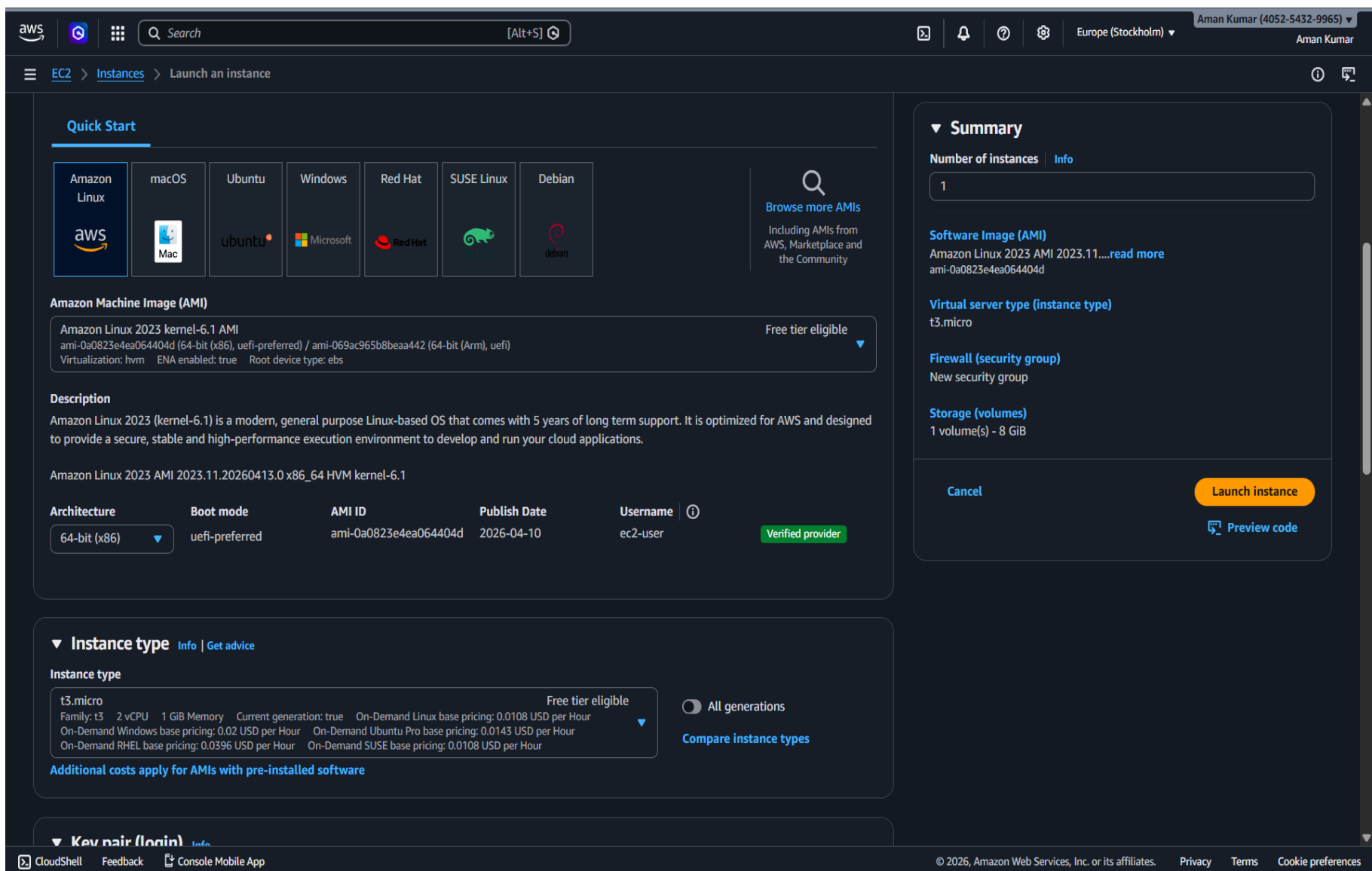
Aim: To launch a virtual machine using AWS EC2 Instance.

Procedure:

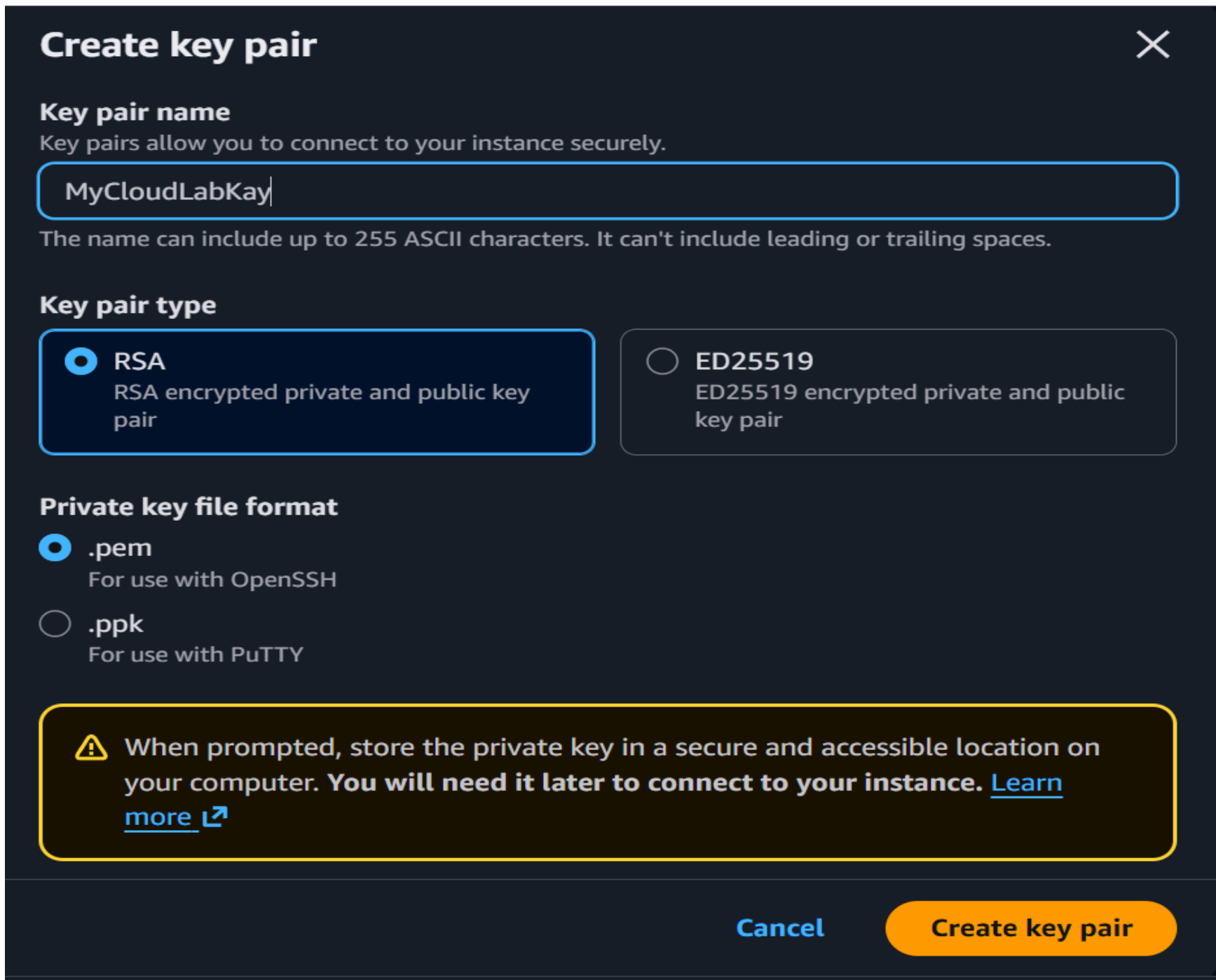
1. Open EC2 dashboard.



2. Select Amazon Linux AMI & t3. micro instance.



3. Create a key pair.



4. Launch instance.

aws

Search

[Alt+S]

Europe (Stockholm)

Aman Kumar (4052-5432-9965)

Aman Kumar

EC2 > Instances

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☒	MyFirstServer	i-06d4f4ccb49355af3	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1b	ec2-13-60-186-181.eu-...	13.60.186.181	-

i-06d4f4ccb49355af3 (MyFirstServer)

DetailsStatus and alarmsMonitoringSecurityNetworkingStorageTags

Instance summary

Instance ID

i-06d4f4ccb49355af3

IPv6 address

-

Hostname type

IP name: ip-172-31-39-41.eu-north-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.60.186.181 [Public IP]

IAM role

-

Public IPv4 address

13.60.186.181 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-39-41.eu-north-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-025f8877872953faa

Subnet ID

-

Private IPv4 addresses

172.31.39.41

Public DNS

ec2-13-60-186-181.eu-north-1.compute.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

5. Connect using EC2 Instance Connect.

aws

Search

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Aman Kumar (4052-5432-9965)

Aman Kumar

```
~$ ssh -i ~/.ssh/instance-connection.pem ec2-user@ip-172-31-39-41
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-39-41 ~]$
```

i-06d4f4ccb49355af3 (MyFirstServer)

PublicIPs: 13.60.186.181 PrivateIPs: 172.31.39.41

CloudShell

Feedback

Console Mobile App

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PRACTICAL 3:

Aim: To create and configure a Virtual Private Cloud (VPC).

Procedure:

1. Create VPC

Create VPC

Resources to create

Info

Create only the VPC resource or the VPC and other networking resources.

VPC only

VPC and more

Name tag - optional

Info

Creates a tag with a key of 'Name' and a value that you specify.

PRACTICAL-VPC

IPv4 CIDR block

Info

IPv4 CIDR manual input

IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/24

CIDR block size must be between /16 and /28.

IPv6 CIDR block

Info

No IPv6 CIDR block

IPAM-allocated IPv6 CIDR block

Amazon-provided IPv6 CIDR block

IPv6 CIDR owned by me

Tenancy

Info

Default

VPC encryption control (\$)

Info

Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. Additional charges apply

None

Monitor mode

Enforce mode

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name

Q PRACTICAL-VPC

Remove tag

Add tag

You can add 49 more tags

Cancel

Preview code

Create VPC

2. VPC resource map for Showing public and private subnets.

aws

Search

[Alt+S]

Europe (Stockholm)

Aman Kumar (4052-5432-9965)

VPC

Your VPCs

VPC dashboard

Filter by VPC

Virtual private cloud

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

Service networks

Lattice services

Resource configurations

Resource gateways

Target groups

Domain verifications

DNS firewall

Rule groups

Your VPCs

VPC encryption controls

Your VPCs (1/2)

Find VPCs by attribute or tag

	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	-	vpc-025f8877872953faa	Available	-	-	Off	172.31.0.0/16	-
☒	Practical-VPC	vpc-071eb77da9796829d	Available	-	-	Off	10.0.0.0/16	-

vpc-071eb77da9796829d / Practical-VPC

Details

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

VPC

Your AWS virtual network

Practical-VPC

Subnets (2)

Subnets within this VPC

eu-north-1a

Public-Subnet

Private-Subnet

Route tables (2)

Route network traffic to resources

Public-RT-Custom

rtb-02fa391e9eaa7105b

Network Connections (1)

Connections to other networks

Practical-IGW

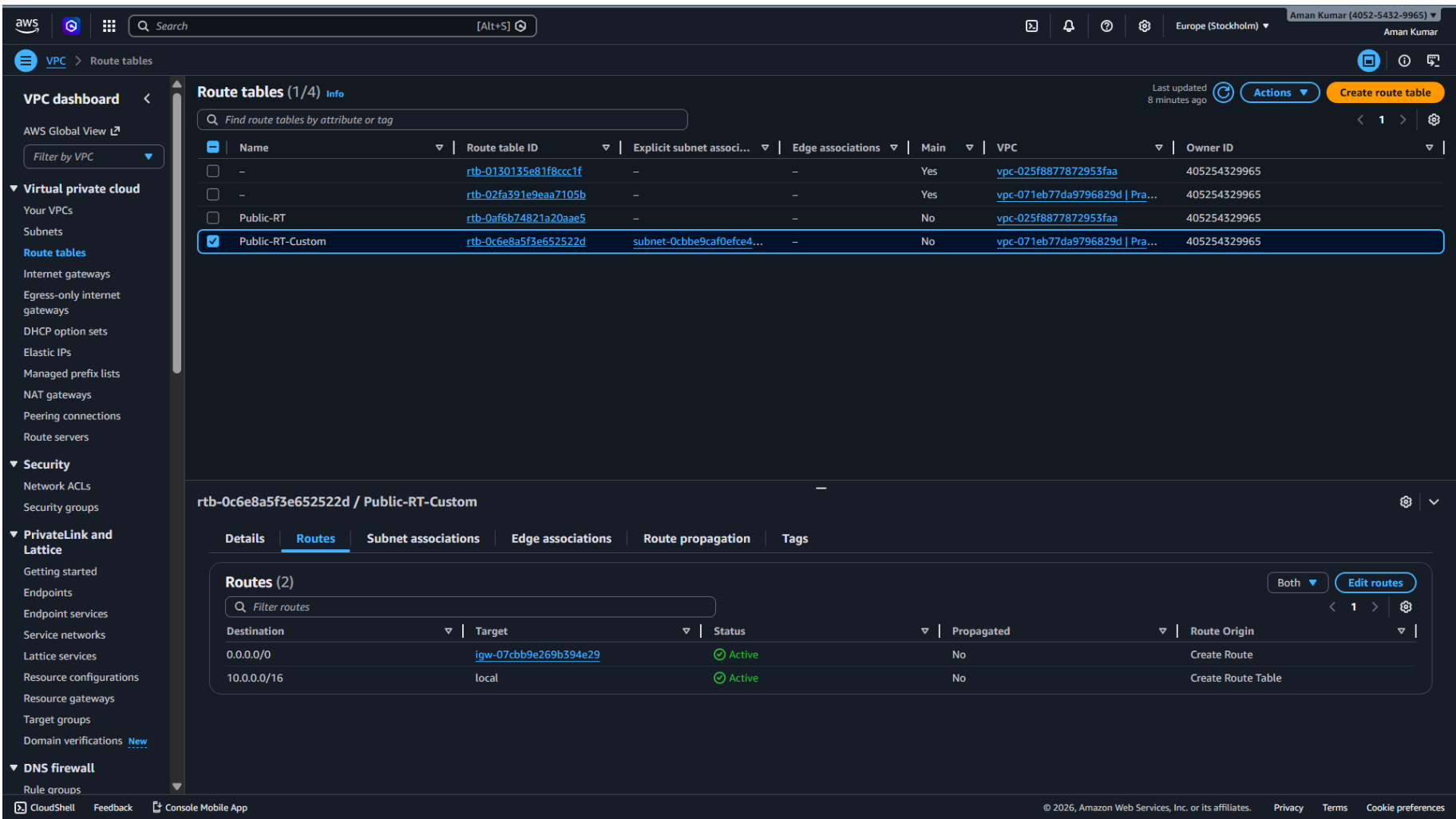
CloudShell

Feedback

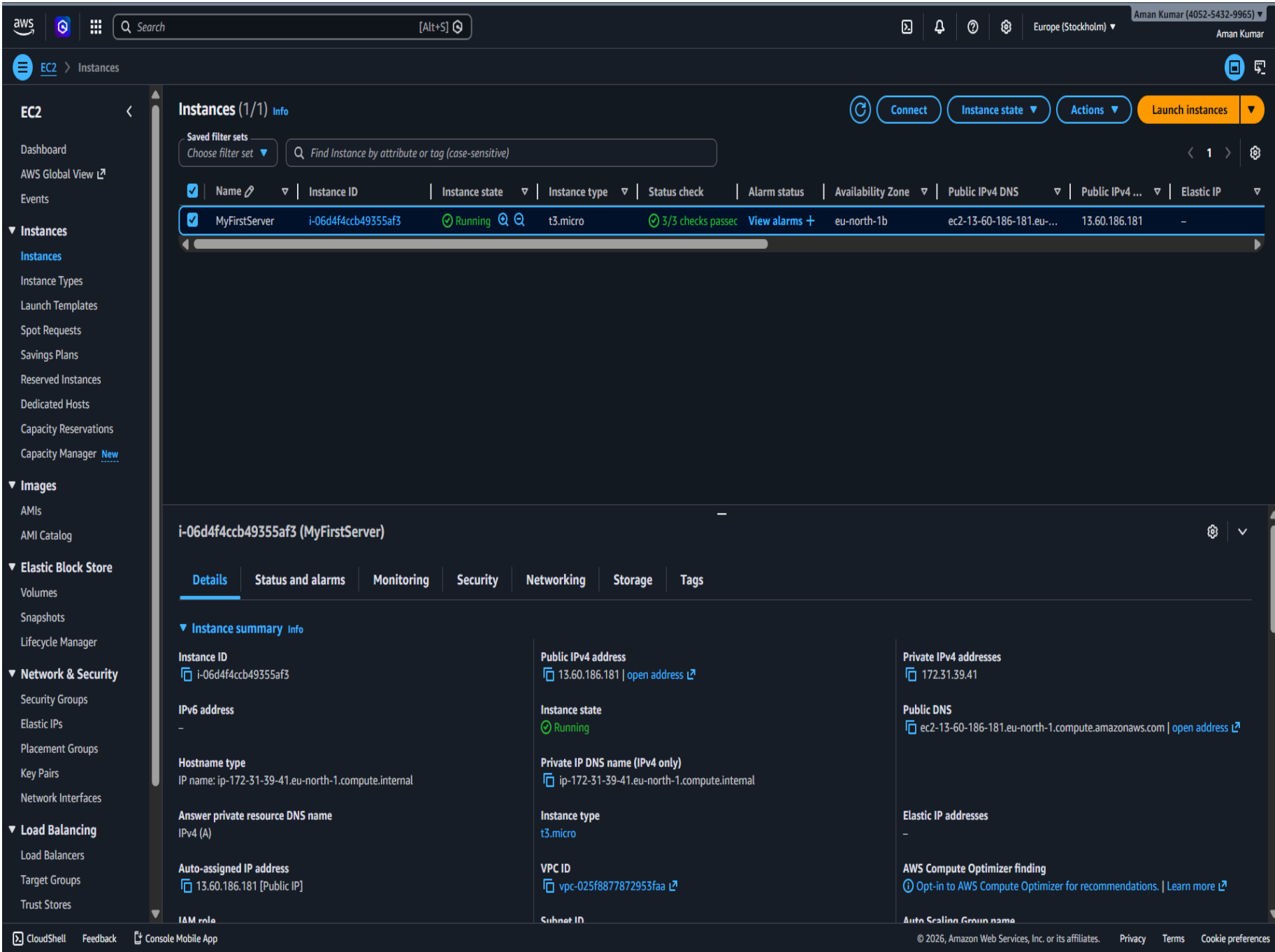
Console Mobile App

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3. Route Table Routes.



4. EC2 Instances View.



2. Select VPC and Subnets.

VPC Lattice integration options [Info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

☒ **No VPC Lattice service**
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ **Attach to VPC Lattice service**
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#) ^U

Application Recovery Controller (ARC) zonal shift - *new* [Info](#)

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ **Enable zonal shift**
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

[Always enabled](#)

Additional health check types - *optional* [Info](#)

☐ **Turn on Elastic Load Balancing health checks** Recommended
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ **Turn on VPC Lattice health checks**
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ **Turn on Amazon EBS health checks**
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

 seconds

Cancel

Skip to review

Previous

Next

3. Set capacity (min=1, max=3).

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

1

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

3

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ No scaling policies

☐ Target tracking scaling policy

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ No policy

For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

☐ Launch before terminating

Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ Terminate and launch

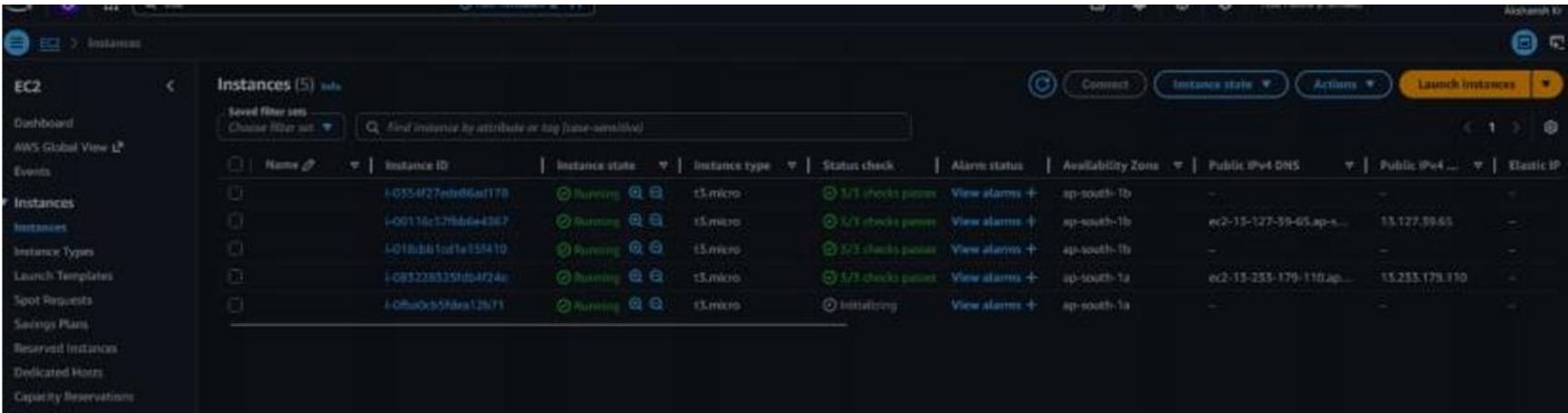
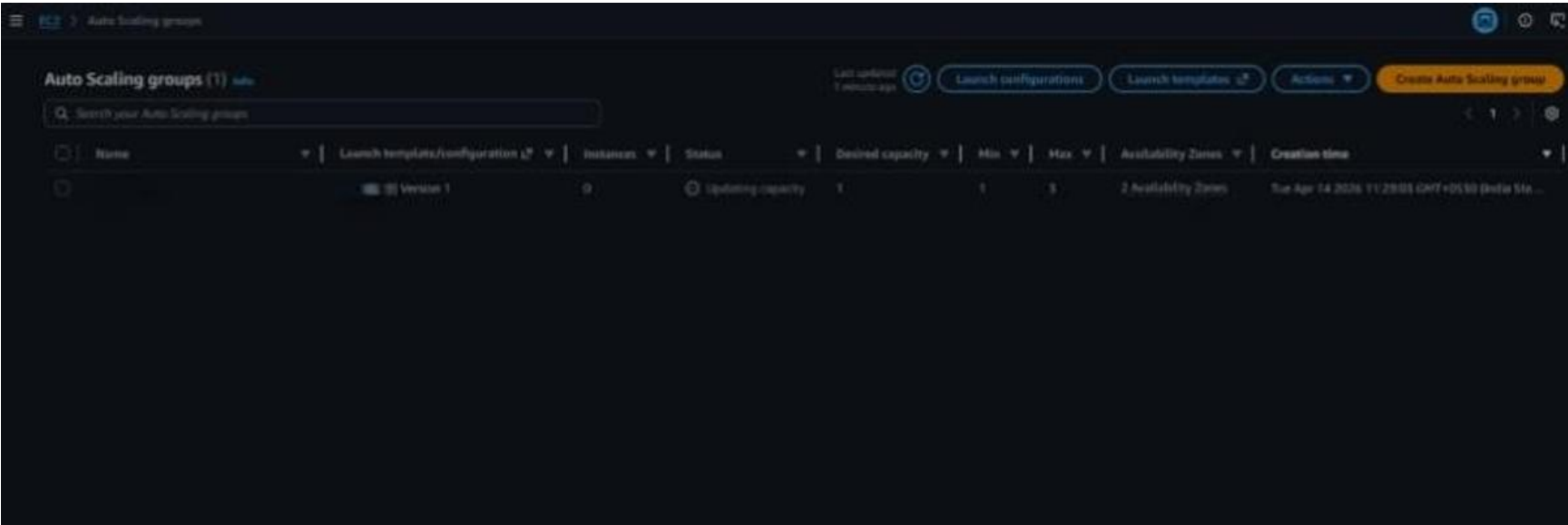
Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexibility

☐ Custom behavior

Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

4.Attach load balancer.

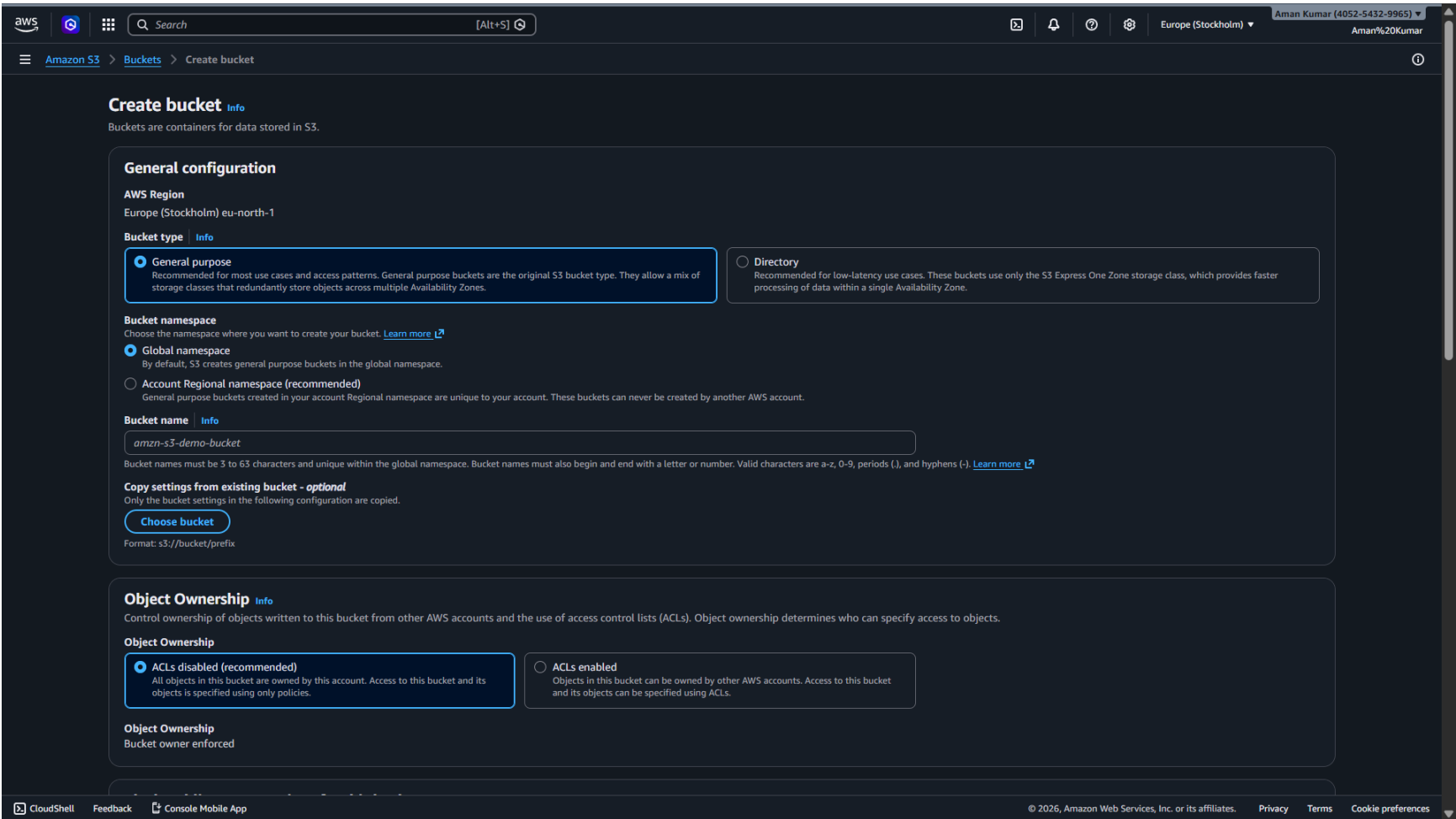


PRACTICAL 5:

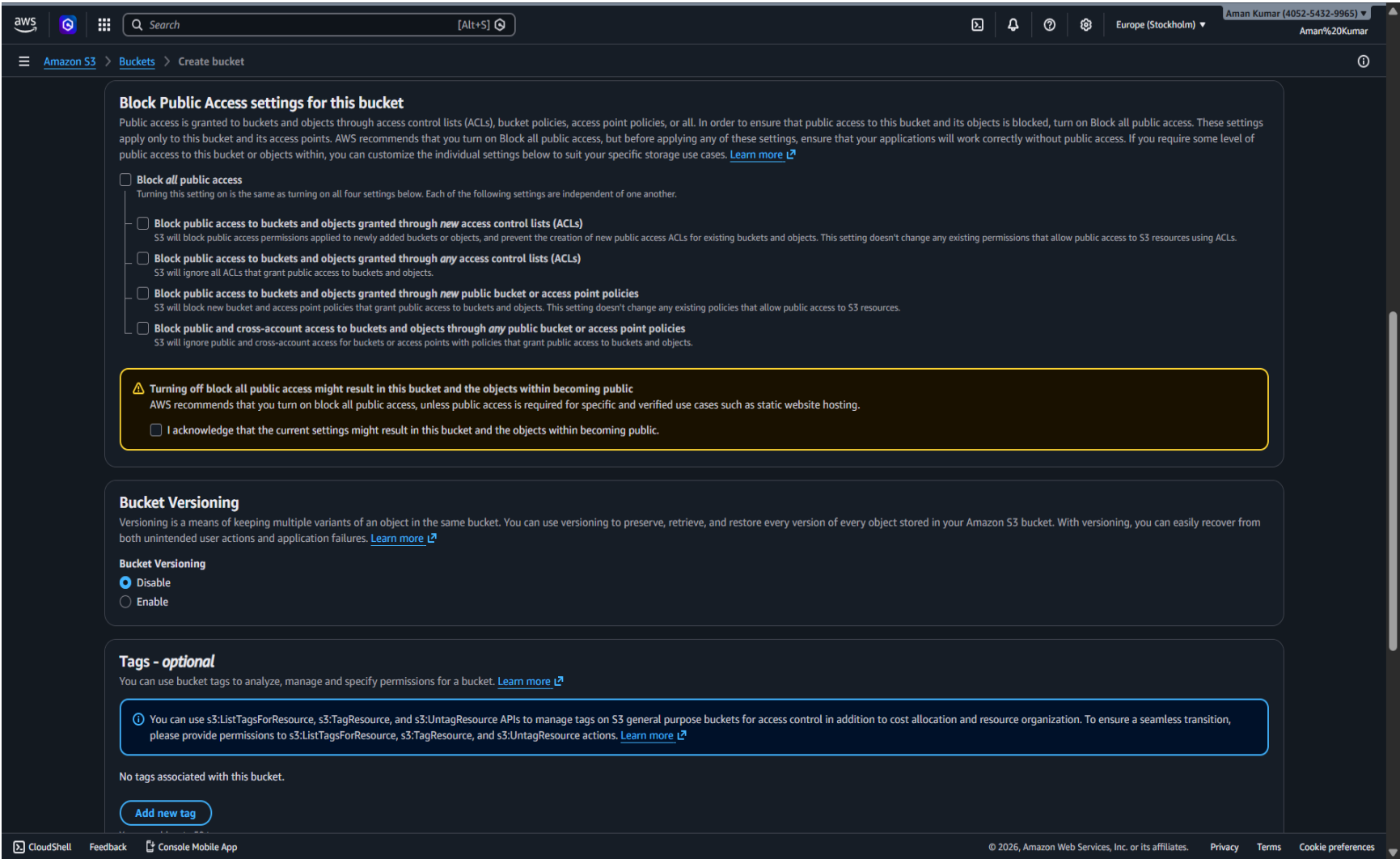
Aim: To host a static website using cloud storage services.

Procedure:

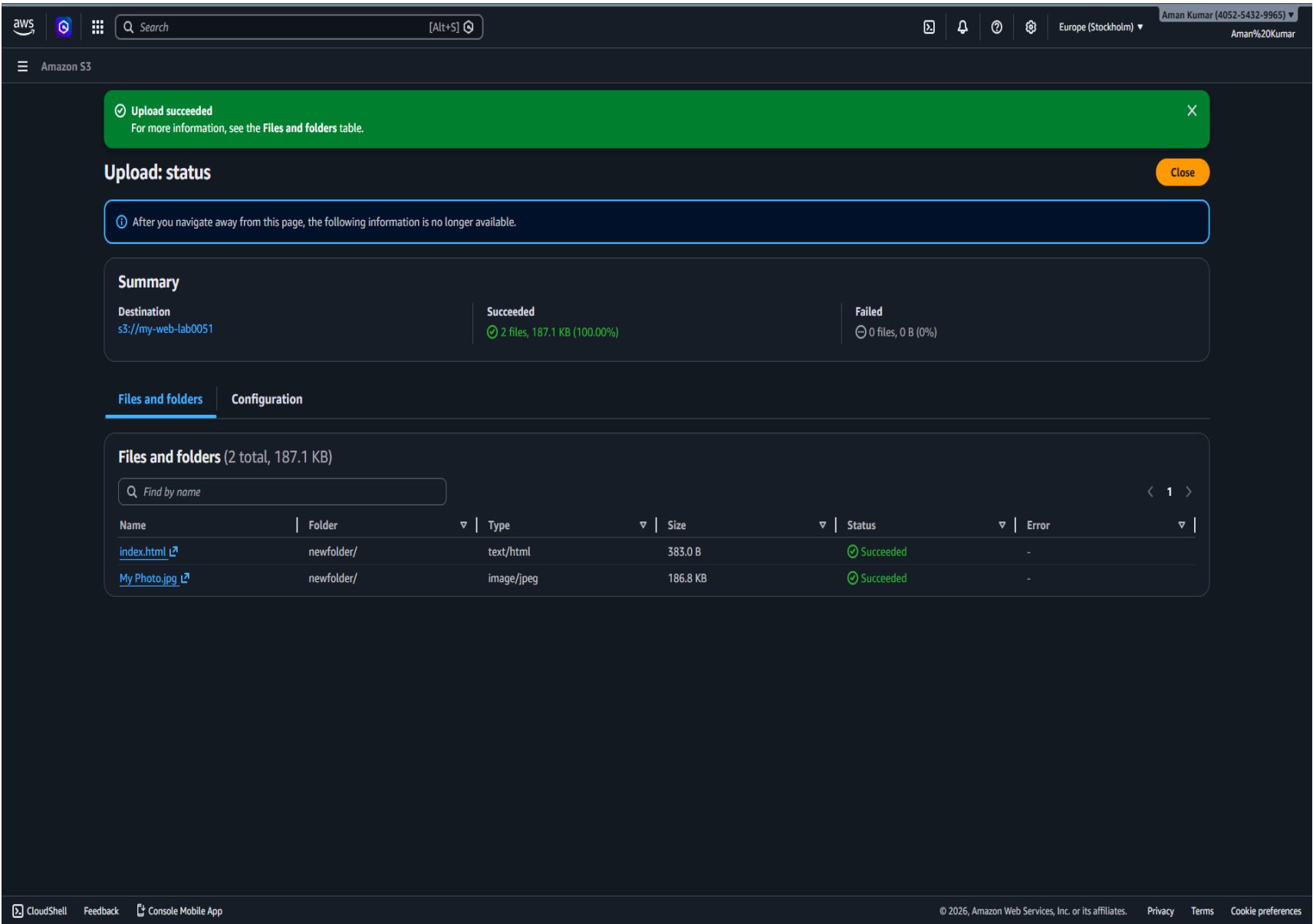
1. Create a S3 bucket.



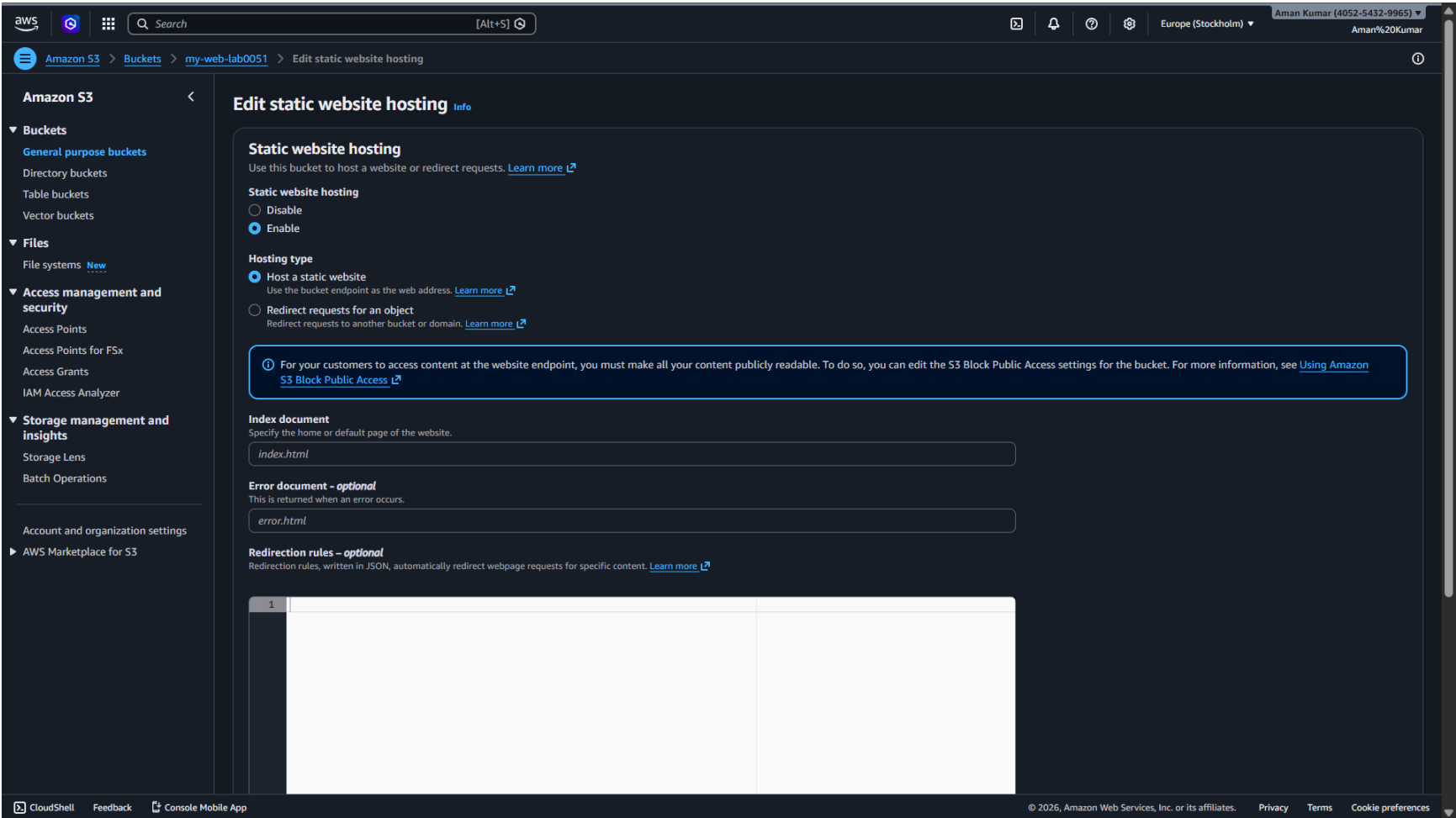
2. Disable the block public access.



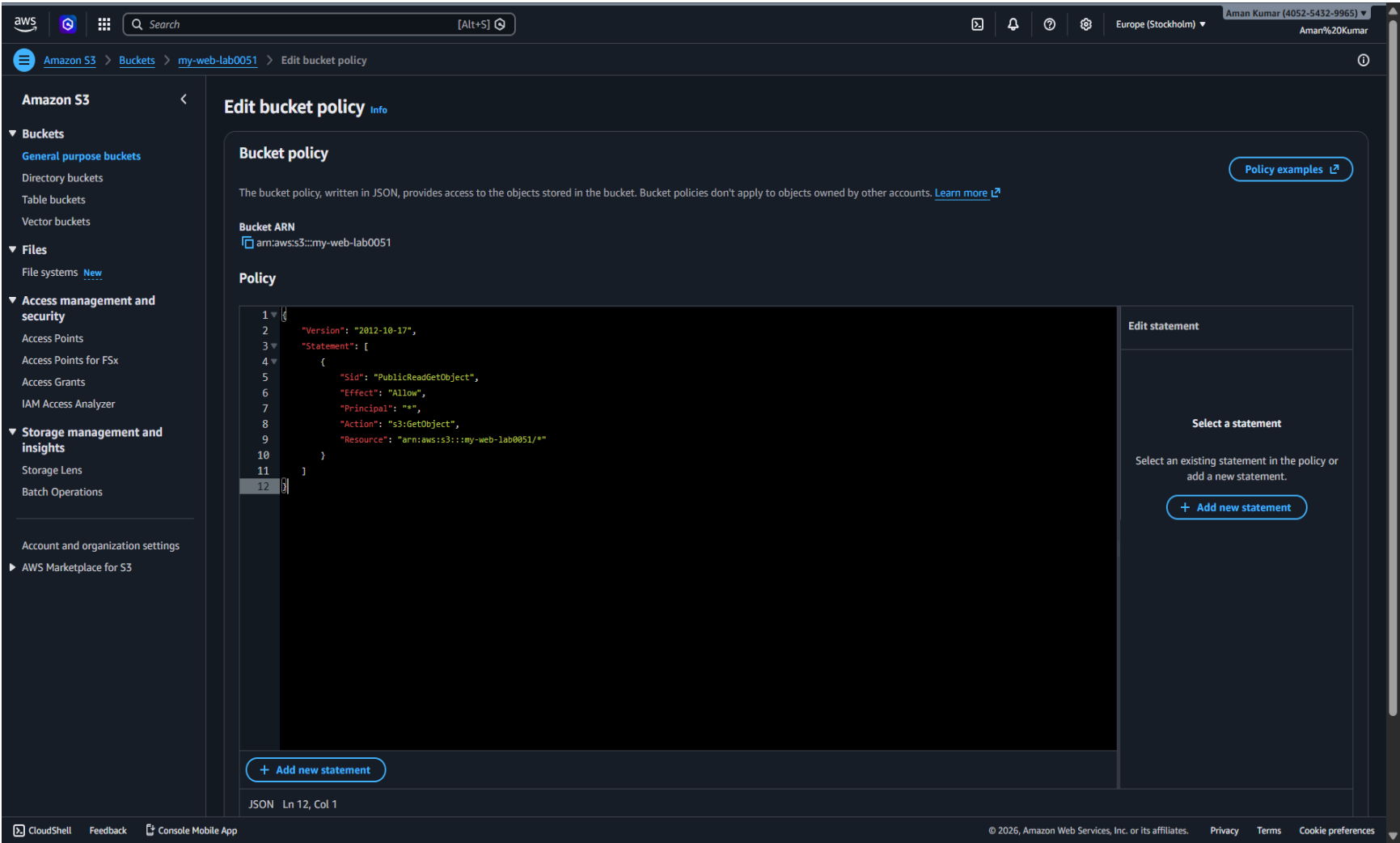
3. Upload the HTML file.



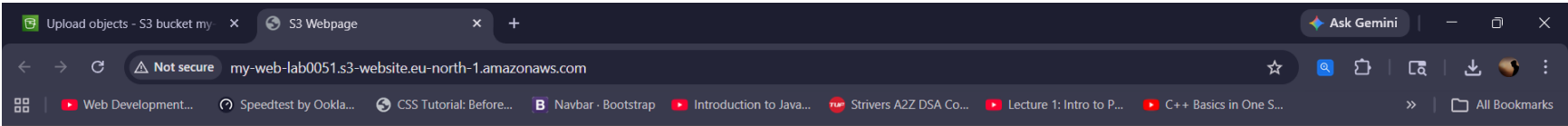
4. Enable static website hosting.



5. Configure permissions.



6. Access the website URL.



Hello! This website is hosted on Amazon S3 by Aman

This is my first S3 webpage.

